Auditing Fairness in ICU Mortality Prediction Across Patient Subgroups

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ABSTRACT

Objective: To evaluate whether machine learning models that predict in-hospital mortality from first-day intensive care unit (ICU) data perform equitably across race, sex and insurance subgroups, and whether bias mitigation reduces any disparities found.
Materials and Methods: We studied the first ICU stay of 65,366 adults in MIMIC-IV. Logistic regression and XGBoost were trained on 14 first-day features and compared with a SOFA-only baseline, using a 60/20/20 split and 5-fold cross-validation. Race, sex and insurance were excluded from model inputs and used to audit test-set AUROC and false-negative rate (FNR) across subgroups, with bootstrap confidence intervals. Reweighting and insurance-specific thresholds were evaluated as mitigation.
Results: XGBoost performed best (AUROC 0.880 vs 0.771 for SOFA baseline). For XGBoost, insurance was the only attribute whose disparities exceeded the 0.05 criterion across the full confidence interval: AUROC gap 0.068 (95% CI 0.051--0.115) and FNR gap 0.138 (0.067--0.308). Reweighting mitigation narrowed the FNR gap from 0.138 to 0.106 with minor change in AUROC, while threshold adjustment widened the gap to 0.186.
Discussion: Insurance-related disparities may reflect differences in socioeconomic circumstances and healthcare access, although causality cannot be inferred. Race-related disparities were inconclusive because small subgroups made gap estimates unstable, and data from single source may limit generalisability.
Conclusion: Strong overall discrimination did not guarantee equitable performance. Neither mitigation strategy achieved the pre-specified 50% reduction in the insurance FNR gap, indicating that more robust approaches are needed.

Key words: algorithmic fairness, intensive care units, hospital mortality, machine learning, MIMIC-IV

INTRODUCTION

In-hospital mortality is a central outcome in intensive care, where timely recognition of patients at high risk of death can guide escalation of treatment and allocation of scarce resources. Machine learning models trained on early physiological data are increasingly proposed for this task, but a model that is accurate on average may still behave unequally across patient groups[1].

Predicting ICU mortality from routinely collected data has a long history in critical-care medicine. Established severity-of-illness scores such as APACHE II[2], SAPS II[3] and the Sequential Organ Failure Assessment (SOFA) score[4] summarise the worst physiological values within the first 24 hours of admission and remain the clinical benchmark against which newer models are judged. The release of large, de-identified datasets, in particular MIMIC-IV[5,6], has made data-driven mortality prediction broadly reproducible, and benchmark feature sets based on first-day summaries are now standard[7,8]. Alongside this progress, concern has grown that clinical prediction models can encode and amplify existing health inequities. Obermeyer et al. showed that a widely deployed risk tool systematically underestimated the needs of Black patients because it used historical cost as a proxy for illness[1], and subsequent work has documented how electronic health record data introduce bias through differential measurement, missingness and coding[9,10]. A model may achieve strong overall discrimination while missing a disproportionate share of deaths in specific subgroups, a failure invisible to aggregate metrics but consequential at the bedside[11,12]. Formal fairness criteria such as equalised odds have been proposed to make such disparities measurable[13].

We ask whether an ICU in-hospital mortality model built from first-24-hour data achieves comparable discrimination and, critically, comparable missed-death (false-negative) rates across race, sex and insurance subgroups, and whether any disparity identified can be reduced by simple fairness mitigation strategies without materially harming overall performance.

MATERIALS AND METHODS

Ethics Statement

This study used MIMIC-IV v3.1, a publicly available, de-identified database accessed under the PhysioNet Credentialed Health Data Use Agreement. No patient-level data generated during the project will be redistributed. Race, sex and insurance are sensitive attributes and were used only for fairness auditing.

Dataset and cohort

We accessed MIMIC-IV v3.1 dataset through Google BigQuery under PhysioNet credentialed access[5,6]. The unit of analysis was a single ICU stay. First-day physiological summaries (vital signs, laboratory values, Glasgow Coma Scale and blood gas) were obtained from the MIMIC-IV derived `first_day_*` concept tables; illness severity was captured by the first-day SOFA score[4] and comorbidity burden by the Charlson Comorbidity Index[14]. The outcome was in-hospital mortality (`hospital_expire_flag`).

Starting from 94,458 ICU stays, we retained adult admissions (age $\geq$ 18) and, because a patient may have several stays, kept each patient's *first* ICU stay ordered by admission time (`intime`), yielding one row per patient. Ordering by admission time rather than the surrogate stay identifier is important, as the latter is not chronological; applying it changed the observed mortality from 12.3% to 10.8%, consistent with later admissions being more severe. The final cohort comprised **65,366 unique adult patients**, of whom **7,086 (10.8%)** died in hospital (Figure 1). The resulting class imbalance (≈9:1) motivates evaluation with AUPRC and subgroup false-negative rates rather than accuracy alone.

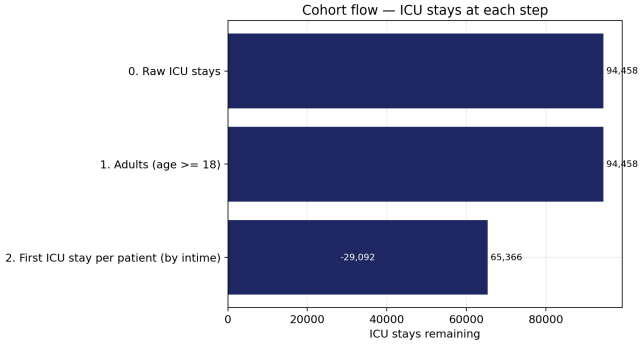


Figure 1. Cohort construction. Of 94,458 adult ICU stays, one first stay per patient (by admission time) was retained, giving 65,366 unique patients.

Sensitive attributes (race, sex and insurance) were excluded from the predictive feature set of the original models but retained for subgroup fairness auditing and mitigation. They were not used as predictors in the SOFA, logistic regression or XGBoost models. Granular MIMIC race labels were collapsed into six categories (White, Black, Hispanic, Asian, Other, Unknown); missing race or insurance was retained as "Unknown" rather than excluded. Most features were near-complete (<1.1% missing); the exception was maximum lactate (45.9% missing), reflecting selective blood-gas sampling, with a 15-percentage-point spread across race groups that we record as differential data quality[9] (Figure 4). Subgroup composition and crude mortality are summarised in Table 1 and Figure 2, with feature-level summary statistics in Table 2. Crude mortality rose monotonically with first-day SOFA score (Figure 3), confirming the expected severity gradient, and no imputation was performed at the cohort stage to prevent leakage.

Table 1. Cohort composition and crude in-hospital mortality by subgroup (n = 65,366; overall mortality 10.8%).

Attribute	Subgroup	n	Mortality (%)
Race	White	42,871	9.5
	Black	6,016	9.1
	Hispanic	2,284	7.5
	Asian	1,981	10.7
	Other	2,833	9.3
	Unknown	9,381	19.3
Sex	Female	28,646	11.3
	Male	36,720	10.5
Insurance	Medicare	34,464	13.1
	Private	18,447	6.9
	Medicaid	9,353	9.1
	Other	1,740	7.2

Table 2. Summary of first-day features (n = 65,366). SD, standard deviation.

Feature	Mean	SD	Median	% miss.
Age (years)	63.4	17.1	65.0	0.0
Heart rate (bpm)	84.1	15.8	82.6	0.1
SBP, min (mmHg)	93.1	17.7	92.0	0.2
SpO_2, min (%)	91.7	6.6	93.0	0.2
Resp. rate (/min)	19.0	3.8	18.4	0.3
Creatinine, max	1.46	1.61	1.00	0.9
Platelets, min	191.0	97.9	178.0	1.1
BUN, max	26.2	21.8	19.0	0.9
WBC, max	14.0	11.7	12.3	1.1
GCS, min	13.7	2.7	15.0	0.4
Lactate, max	3.07	2.82	2.20	45.9
SOFA score	4.24	3.35	4.00	0.0
Charlson index	4.66	3.01	4.00	0.0

Machine learning models and evaluation

Data were split into stratified training (60%, n = 39,219), validation (20%, n = 13,073) and test (20%,

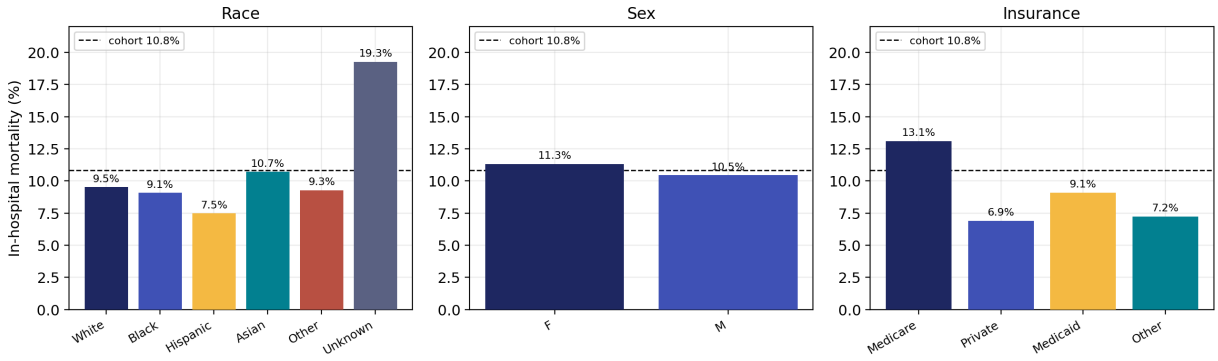


Figure 2. Crude in-hospital mortality by race, sex and insurance. The dashed line marks the cohort average (10.8%). The Unknown-race group (19.3%) stands well above all others.

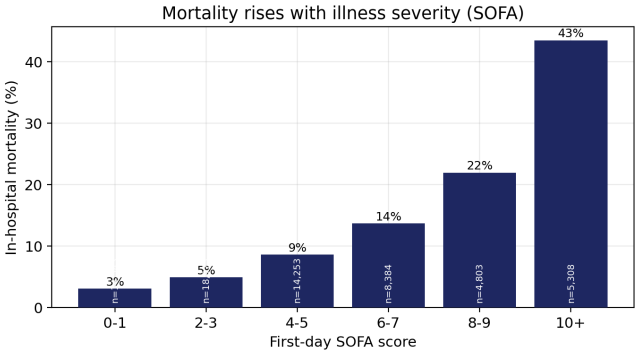


Figure 3. In-hospital mortality by first-day SOFA score band, confirming the expected severity gradient.

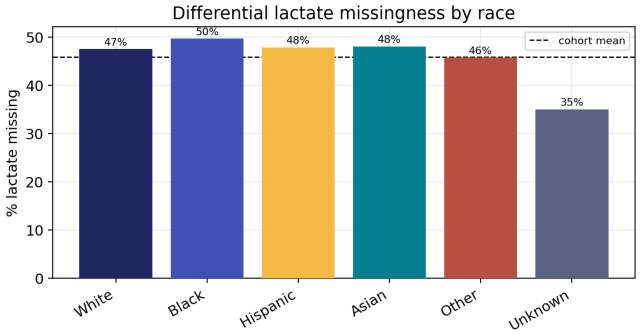


Figure 4. First-day lactate missingness by race (cohort mean 45.9%), illustrating differential data quality across subgroups.

$n = 13{,}074$) sets, each with 10.8% mortality. The validation set was excluded from model fitting and reserved for threshold selection in the fairness audit. The test set was reserved for subgroup fairness auditing and final performance evaluation and was not used for model fitting or threshold tuning. A lactate-measured indicator was added (45.9% of lactate values were missing), giving 14 features. Missing values were median-imputed and features standardised within a pipeline fitted on training data only. Race, sex and insurance were excluded from all models.

Three models were compared, using two of the most common methods for ICU mortality prediction[15]: (1) baseline logistic regression on SOFA alone, (2) logistic regression and (3) XGBoost, both on all 14 features. Hyperparameters were tuned by stratified 5-fold cross-validation on the training set (logistic regression: $C = 0.01$; XGBoost, 15-configuration randomised search: 800 trees, depth 4, learning rate 0.03, row and column subsampling 0.8). Class weighting was not used since it inflated mean predicted risk on the validation set (31% for XGBoost vs 10.8% actual mortality) without improving discrimination.

The primary metric was the area under the ROC curve (AUROC), a standard discrimination measure for clinical prediction models[16]; unlike AUPRC, it does not favour subgroups with higher mortality[17]. The area under the precision-recall curve (AUPRC), which focuses on deaths[18], was the tuning objective and a secondary metric. The Brier score summarised the accuracy of predicted probabilities. Models were implemented in scikit-learn and XGBoost. Logistic regression coefficients and SHAP values[19] were used to interpret feature effects.

Fairness audit

We conducted a post-hoc fairness audit across three patient attributes: race, sex, and insurance status. Missing race and insurance values were retained as an "Unknown" category to avoid excluding patients with incomplete demographic information.

Subgroup performance was evaluated on the held-out test set using AUROC, AUPRC, false negative rate (FNR), false positive rate (FPR), and Brier score. FNR was defined as the proportion of actual deaths incorrectly classified as survivors. For each model, a decision

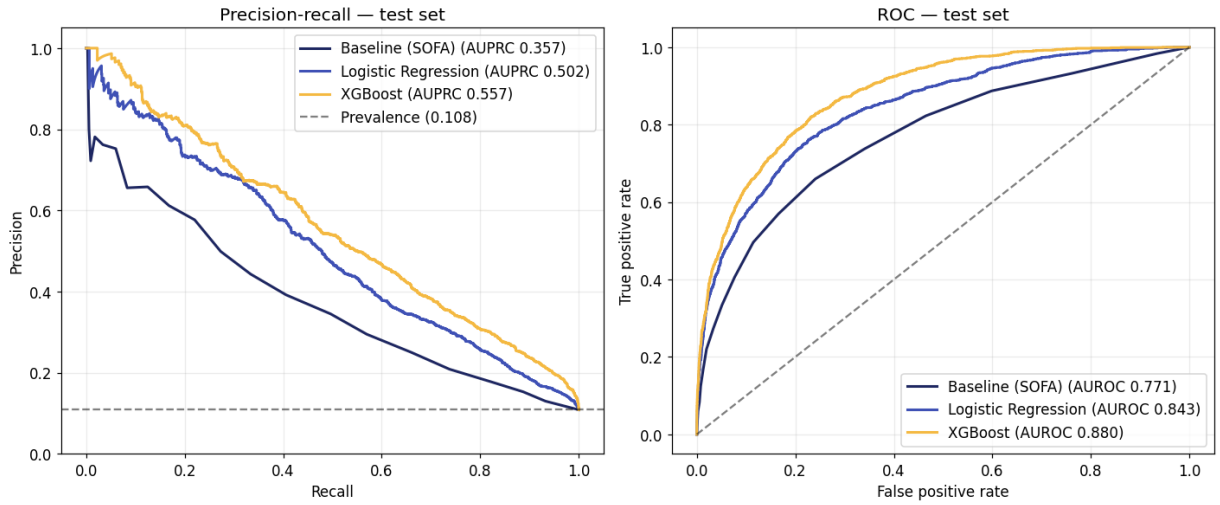


Figure 5. Precision-recall (left) and ROC (right) curves on the held-out test set ($n = 13{,}074$). The dashed line is the mortality prevalence (left) or chance performance (right).

threshold targeting 80% sensitivity was selected on the validation set and applied unchanged to all test subgroups.

Fairness gaps were calculated as the difference between the maximum and minimum subgroup values for each metric. Equalised odds difference was defined as the maximum of the true positive rate and false positive rate gaps[13]. Following the pre-specified criteria, an AUROC gap exceeding 0.05 or an FNR gap exceeding five percentage points indicated a fairness concern.

Uncertainty was assessed using 1,000 bootstrap resamples stratified by subgroup, with a fixed random seed of 42. Percentile-based 95% confidence intervals were calculated for fairness gaps[20]. Subgroups with fewer than 500 patients were flagged as unstable, consistent with the proposal. Fewer than 50 observed deaths was used as an additional stability flag. Flagged groups were retained but interpreted cautiously.

Bias mitigation

Two mitigation strategies were evaluated for the primary insurance-related disparity. First, reweighting was applied during training by assigning sample weights to each insurance–mortality combination so that insurance and mortality were approximately independent in the weighted training distribution[21,22]. Second, insurance-specific threshold adjustment was applied as a post-processing strategy. Separate decision thresholds were selected on the validation set to target approximately 80% sensitivity within each insurance group, with the aim of reducing between-group false-negative-rate differences[13]. Both strategies were evaluated on the unchanged held-out test set using overall AUROC, maximum FNR gap, and equalized-odds (EO) difference. Mitigation was considered successful if the primary insurance FNR gap was reduced by at least 50% while the overall AUROC decreased by less than 0.02 relative to the unmitigated model.

RESULTS

Overall model performance

The model performance results in Table 3 and Figure 5 showed that each additional level of model complexity improved AUROC, AUPRC and Brier score, with XGBoost curves highest across almost the entire range. On the test set, XGBoost achieved an AUROC of 0.880 and AUPRC of 0.557, compared with 0.843 and 0.502 for logistic regression and 0.771 and 0.357 for the SOFA baseline. Both logistic regression and XGBoost met the pre-specified AUROC target of 0.80. Most of the gain came from adding clinical features rather than from the more complex model (AUROC +0.072 vs +0.037). Test and cross-validation AUPRC were similar, suggesting no overfitting. All Brier scores were below 0.097, the score from predicting the prevalence for everyone.

Table 3. Overall ML models performance on the test set ($n = 13{,}074$).

Model	AUROC	AUPRC	Brier	CV AUPRC
SOFA	0.771	0.357	0.082	0.341 ± 0.022
LR	0.843	0.502	0.072	0.493 ± 0.022
XGBoost	**0.880**	**0.557**	**0.068**	$\mathbf{0.541 \pm 0.023}$

LR, logistic regression; CV AUPRC, mean ± SD over five training folds.

Logistic regression and XGBoost both ranked SOFA, Charlson index and respiratory rate as the strongest pre-

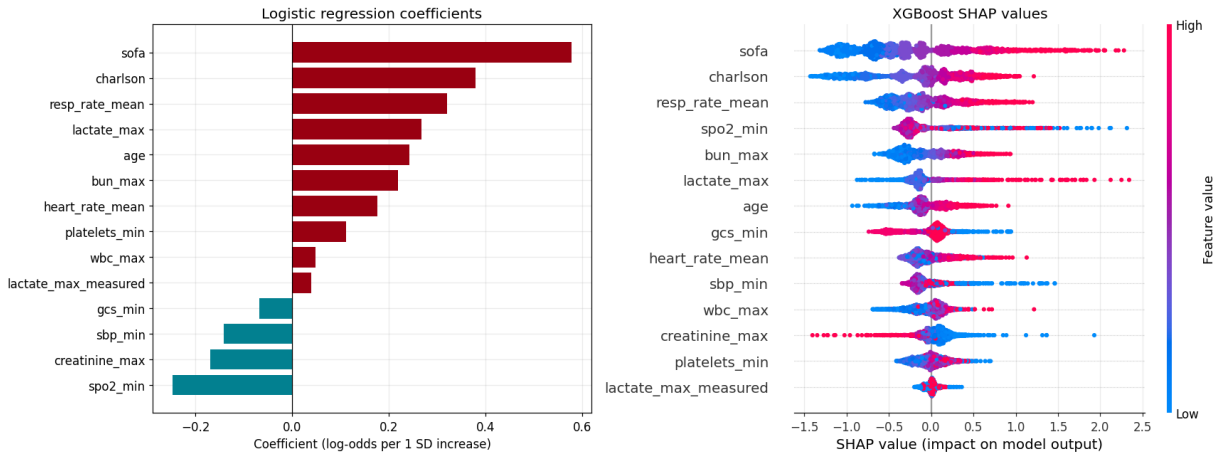


Figure 6. Feature effects. Left: logistic regression coefficients (change in log-odds per 1 SD increase). Right: SHAP values for XGBoost on 2,000 test patients; each point is a patient, colour shows the feature value, and positive values increase predicted risk.

dictors, with consistent directions: higher SOFA, comorbidity, lactate and age, and lower SpO_2, GCS and blood pressure, increased predicted risk (Figure 6). For SpO_2, blood pressure and lactate, most patients had SHAP values near zero, but extreme values formed a long tail of higher risk. Logistic regression assumes linear effects and cannot capture such thresholds, which may partly explain XGBoost's better performance. XGBoost was therefore carried forward as the primary model for bias mitigation.

Subgroup fairness audit

Insurance status exhibited the largest fairness disparities for both XGBoost and logistic regression (LR). For XGBoost, the insurance AUROC gap was 0.0682 (95% CI: 0.0506–0.1148), and the FNR gap was 0.1378 (95% CI: 0.0673–0.3077). LR showed larger corresponding gaps of 0.1020 (95% CI: 0.0600–0.1506) and 0.2041 (95% CI: 0.1199–0.3651). All four confidence intervals had lower bounds exceeding the pre-specified threshold of 0.05.

For XGBoost, race showed an FNR gap of 0.0757 (95% CI: 0.0493–0.2835), exceeding the criterion at the point-estimate level but not across the full confidence interval. The Asian subgroup had the highest FNR (0.250), although only 44 deaths were observed, limiting estimate stability. Sex-related disparities were smaller, with XGBoost AUROC and FNR gaps of 0.0113 and 0.0136, respectively.

Although XGBoost achieved better overall discrimination than LR, subgroup disparities remained. Insurance was therefore prioritised for the subsequent mitigation analysis. These findings are summarised in Table 4 and Figure 7.

Table 4. Fairness gaps across patient attributes. Values represent the maximum difference between subgroups.

Model	Attribute	AUROC gap	FNR gap
XGBoost	Race	0.0387	0.0757
	Sex	0.0113	0.0136
	Insurance	0.0682	0.1378
LR	Race	0.0494	0.0286
	Sex	0.0146	0.0243
	Insurance	0.1020	0.2041

A gap exceeding 0.05 indicates a fairness concern under the pre-specified criteria.

Mitigation results

As shown in Table 5, the unmitigated XGBoost model achieved an overall AUROC of 0.880, with a maximum insurance FNR gap of 0.138 and an EO difference of 0.153. Reweighting produced a partial improvement in error-rate fairness: the FNR gap decreased to 0.106, corresponding to a 23.4% relative reduction, while the EO difference decreased to 0.135. Overall discrimination was essentially preserved, with AUROC decreasing only from 0.880 to 0.879. This pattern is consistent with previous evidence that reweighting can reduce subgroup disparities while retaining predictive performance, although the magnitude of benefit may vary across models and fairness measures[21,22]. However, the improvement did not meet the pre-specified mitigation criterion of at least a 50% reduction in the primary FNR gap.

In contrast, threshold adjustment did not improve fairness on the held-out test set. The maximum FNR gap increased from 0.138 to 0.186, while the EO difference increased from 0.153 to 0.202. AUROC remained 0.880, because threshold adjustment changes classification decisions but not the ranking of predicted

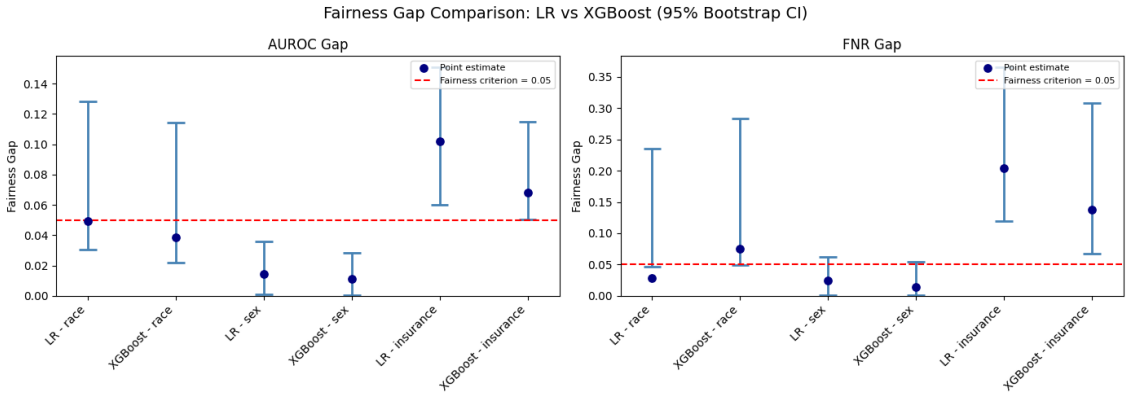


Figure 7. Fairness gaps across race, sex, and insurance groups for logistic regression and XGBoost. Points represent observed gaps and vertical bars show 95% bootstrap confidence intervals. The dashed red line indicates the pre-specified fairness criterion of 0.05.

Table 5. Effect of mitigation on overall performance and fairness gaps.

Method	AUROC	Max FNR gap	EO difference
No mitigation	0.880	0.138	0.153
Reweighting	0.879	0.106	0.135
Threshold adjustment	0.880	0.186	0.202

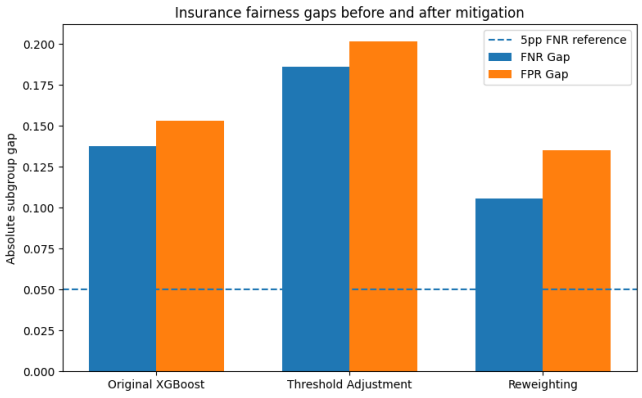


Figure 8. Insurance-related FNR and FPR gaps before and after mitigation. Reweighting reduced both gaps relative to the original XGBoost model, whereas threshold adjustment increased them. The dashed line marks the pre-specified 5 percentage-point FNR fairness criterion.

risks. Therefore, reweighting provided a modest fairness improvement with minimal performance loss, whereas threshold adjustment worsened the observed insurance disparity. These mitigation results are summarised visually in Figure 8, which compares the insurance-related FNR and FPR gaps before and after threshold adjustment and reweighting.

DISCUSSION AND CONCLUSION

The fairness audit demonstrated that stronger overall predictive performance does not necessarily ensure equitable outcomes across patient groups. Despite achieving higher discrimination, XGBoost retained substantial insurance-related disparities in false negative rates, which may have clinical implications if predictions inform patient monitoring or resource allocation.

Insurance status may reflect differences in socioeconomic circumstances and healthcare access, although these observational results do not establish causality. Such disparities are consistent with concerns that clinical algorithms may reproduce existing healthcare inequalities[1,11].

Race-related disparities were less conclusive due to small subgroup event counts. We therefore prioritised insurance for mitigation while recognising the need to monitor fairness across other demographic groups.

Several limitations should be considered. First, MIMIC-IV is derived from a single academic medical centre, which may limit generalisability to other hospitals and patient populations. Second, race and insurance were based on routinely recorded demographic and administrative categories, with missing values retained as an "Unknown" group; these categories may not fully capture the underlying social and socioeconomic factors relevant to model performance. Third, lactate showed substantial missingness (45.9%), with missingness varying across racial groups, which may introduce measurement-related bias. Fourth, fairness estimates for smaller subgroups or those with few deaths were less stable, and the observational design does not establish that insurance status itself causes the observed disparities. Because insurance was prioritised for mitigation after observing the subgroup audit on the held-out test set, the mitigation analysis should be considered exploratory. Future

studies should pre-specify the mitigation target or confirm mitigation performance in an independent external dataset.

Overall, stronger predictive discrimination did not guarantee equitable model performance across patient subgroups. XGBoost achieved the best overall discrimination (AUROC 0.880) but showed the clearest disparities across insurance groups. Reweighting reduced the insurance FNR gap from 0.138 to 0.106 with negligible loss in AUROC, although it did not meet the prespecified 50% reduction target. In contrast, insurance-specific threshold adjustment increased both FNR and equalised-odds disparities on the held-out test set. These findings highlight the importance of evaluating subgroup fairness alongside conventional predictive performance. Future work should validate these findings in external, multi-centre populations and assess more robust mitigation strategies.

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GENERATIVE AI USE DECLARATION

We used ChatGPT and Claude to improve sentence clarity, to help debug code, and to support group meetings during planning of the project. All AI outputs were reviewed and revised by the authors before inclusion. The final research question, data preprocessing, feature engineering, model implementation, experiments, figures, tables and interpretations were developed and verified by the authors. No MIMIC-IV patient-level data were shared with any AI tool. No AI-generated data, references or experimental results were used. A full record of prompts and outputs is available upon request.

AUTHOR CONTRIBUTIONS

Table 6. Author contributions using the CRediT (Contributor Roles Taxonomy).

Author	CRediT roles
Rohan Roy Mathew	Conceptualization (study design and research question), Methodology (cohort definition and digital phenotyping), Software (Part A – data extraction and cohort construction), Data curation (MIMIC-IV extraction, cohort deduplication, missingness analysis), Formal analysis (cohort characterisation and subgroup mortality), Visualization (Figure 1, Figure 2, Figure 3, Figure 4, Table 1, Table 2), Writing – original draft (Introduction; Methods: dataset and cohort), Writing – review & editing
Ya-Chu Yang	Methodology (machine learning models and evaluation), Software (Part B - model training and evaluation), Formal analysis (model performance and interpretability), Validation (overall model performance), Visualization (Figure 5, Figure 6, Table 3), Writing – original draft (abstract; Methods: ML and evaluation; Results: overall performance), Writing – review & editing
Yihan Shi	Methodology (fairness audit design and subgroup evaluation), Software (Part C - fairness audit implementation), Formal analysis (subgroup performance disparities in AUROC and FNR, including confidence intervals), Validation (statistical assessment of fairness disparities), Visualization (Table 4, Figure 7), Writing – original draft (Methods: fairness audit; Results: subgroup fairness analysis), Writing – review & editing
Haokun Yang	Methodology (fairness mitigation design), Software (Part D mitigation implementation and evaluation), Formal analysis (before–after comparison of mitigation performance), Validation (mitigation performance), Visualization (Table 5, Figure 8), Writing – (Methods: bias mitigation; Results: mitigation results; Discussion, limitations and conclusion)